

Lesson 3: Geometric Sequences & Series — Practice Worksheet

Name: _____ Date: _____

Attempt every question. Show your working. The sections increase in difficulty.

Section 1 — Warm-up (foundation)

1. Write the first four terms of the geometric sequence with $u_1 = 3$, $r = 2$.
2. A GP has $u_1 = 2$, $r = 3$. Find u_6 .
3. Find the common ratio of 5, 15, 45, 135, ...
4. Find the 5th term of 4, 8, 16, ...
5. Find r for the sequence 80, 40, 20, 10, ...
6. A GP has $u_1 = 1$, $r = -2$. Find u_6 (watch the sign).

Section 2 — Core practice

7. Find the sum of the first 7 terms of 2, 6, 18, ...
8. Find S_8 for the GP with $u_1 = 5$, $r = 2$.
9. Find the sum of the first 5 terms of 3, 12, 48, ...
10. A GP has $u_2 = 6$ and $u_5 = 162$. Find r and u_1 .
11. Find the sum of the first 5 terms of 64, 32, 16, ...

Section 3 — Challenge & exam-style

- 12.** £1500 is invested at 4% per year, compounded annually. Find its value after 8 years (2 d.p.).
- 13.** A car worth £20000 loses 15% of its value each year. Find its value after 3 years (nearest £).
- 14.** The 2nd term of a GP is 18 and the 5th term is 486. Find r and u_1 .
- 15.** A population of 5000 grows by 8% each year. Write a formula for the population after n years, and find it after 10 years (nearest whole).